Constructing a Hadamard Matrix of Order 668: A Computational Investigation via the Goethals–Seidel Array

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Abstract

The Hadamard conjecture asserts that a Hadamard matrix of order n exists for every positive integer n divisible by 4. Order 668 is the smallest multiple of 4 for which no Hadamard matrix construction is known, as confirmed by the Cati–Pasechnik database [6]. We investigate the feasibility of constructing $H(668)$ via the Goethals–Seidel array [17], which reduces the problem to finding four $\{\pm 1\}$-sequences of length 167 whose summed periodic autocorrelation function (PAF) vanishes at every non-zero shift. We establish a Legendre-symbol baseline achieving a uniform PAF of -4 across all non-zero shifts—a gap of exactly 4 from the target—and prove that this configuration is a strict local minimum under single-position perturbations. We deploy an extensive suite of optimization methods including PAF-direct simulated annealing with Numba-accelerated integer arithmetic, parallel tempering, multi-flip combinatorial search, Williamson-type symmetric restrictions, skew-plus-symmetric sequence restrictions, row-sum-targeted initialization, submatrix extraction from $H(672)$, MILP-based flip optimization, mixed algebraic constructions using product sequences, and inter-sequence swap moves, totalling approximately 3.9×10^9 candidate evaluations across 12 experiments. No exact solution was found. PAF-direct simulated annealing achieved a best L_2 cost of 1,984 (versus the Legendre baseline's 2,656) with $L_\infty = 8$ (versus the baseline's $L_\infty = 4$). We analyze the structural barriers posed by the sparse multiplicative group $\mathbb{Z}_{167}^*$ of order $166 = 2 \times 83$ and contextualize our findings against Eliahou's 64-modular result [16]. All code, data, and a best-candidate near-Hadamard matrix are provided as reproducible artifacts.

Keywords: Hadamard matrix, Goethals–Seidel array, supplementary difference sets, simulated annealing, periodic autocorrelation function, order 668

1 Introduction

A *Hadamard matrix* of order n is an $n \times n$ matrix H with entries in $\{+1, -1\}$ satisfying

$$HH^T = nI_n, \tag{1}$$

where I_n is the identity matrix. A necessary condition for the existence of such a matrix is that $n = 1$, $n = 2$, or $n \equiv 0 \pmod 4$. The celebrated *Hadamard conjecture*, open since Hadamard's original investigation [18], asserts that this necessary condition is also sufficient.

Despite extensive progress spanning more than a century—from Sylvester's Kronecker product constructions [33] through Paley's quadratic residue method [24] to modern computational searches [3, 13]—the conjecture remains unresolved. As of the comprehensive database maintained by Cati and Pasechnik [5, 6], **order 668 is the smallest multiple of 4 for which no Hadamard matrix construction is known**. This makes $H(668)$ the current frontier of the conjecture for explicit constructions.

The factorization $668 = 4 \times 167$, where 167 is prime with $167 \equiv 3 \pmod 4$, places severe constraints on applicable construction methods. In this paper we:

1. Systematically eliminate all classical algebraic constructions (Paley I/II, Kronecker, Miyamoto, Williamson) for order 668 (Section 3);
2. Establish a Legendre-symbol baseline with a uniform PAF gap of 4 and prove it is a strict local minimum (Section 4.1);
3. Deploy twelve distinct optimization strategies within the Goethals–Seidel framework, totalling $\sim 3.9 \times 10^9$ evaluations (Section 4);
4. Document the convergence barriers and analyze the structural obstacles posed by the group $\mathbb{Z}_{167}^*$ (Section 6 and Section 7);
5. Provide concrete recommendations for future work (Section 8).

2 Related Work

2.1 Classical Constructions

The theory of Hadamard matrices originates with Sylvester [33], who constructed matrices of all orders 2^k via the Kronecker product. Hadamard [18] proved the determinant bound and noted that $\{\pm 1\}$-matrices achieving it satisfy (1). Paley [24] introduced the quadratic residue constructions yielding infinite families at prime-power-related orders.

Williamson [37] showed that four symmetric circulant $\{\pm 1\}$-matrices satisfying $A^2 + B^2 + C^2 + D^2 = 4nI$ produce a Hadamard matrix of order $4n$. Goethals and Seidel [17] generalized this with their array construction, dropping the symmetry requirement. The Baumert–Hall construction [1] and Turyn's work [35, 36] on T-sequences further extended the toolkit. The Cooper–Wallis construction [8] combined T-matrices with Williamson matrices for additional orders.

Comprehensive treatments appear in the monographs by Horadam [19], Seberry [26], Seberry and Yamada [27, 28], de Launey and Flannery [10], and the Handbook of Combinatorial Designs [7].

2.2 Computational Searches

Djoković and Kotsireas [12–14] conducted extensive computational searches for supplementary difference sets (SDS) and Goethals–Seidel difference families, resolving many previously open orders. Bright, Djoković, Kotsireas, and Ganesh [3, 4] developed a SAT+CAS hybrid framework that enumerates best matrices up to order 208. Kharaghani and Tayfeh-Rezaie [20] constructed $H(428)$, resolving a long-standing open case.

Suksmono [30] explored simulated quantum annealing for Hadamard matrix search, and Suksmono and Minato [32] extended this to hardware quantum annealers. Suksmono [31] further investigated QAOA-based approaches. Bennett et al. [2] recently used quaternionic perfect sequences to enumerate Williamson-type matrices. Shen and Zhang [29] developed a k-block decomposition method for constructing GS sequences with symmetry, and London and Kotsireas [22] constructed the first Turyn-type sequences of lengths 40, 42, and 44. Djoković [15] constructed symmetric Hadamard matrices of order $q(1+q)$ for prime powers $q \equiv 3 \pmod 8$ via orthogonal designs.

2.3 Recent Progress on Order 668

Eliahou [16] achieved a significant partial result: a *64-modular Hadamard matrix* of order 668, i.e., a $\{\pm 1\}$-matrix H with $HH^T \equiv 668I \pmod{64}$. This establishes that modular obstructions (at least for the prime 2 up to the sixth power) do not prevent existence.

The Cati–Pasechnik database [6] confirms 668 as the smallest open case, with the SageMath implementation [34] lacking a construction at this order. The PlanetMath listing [25] identifies 167 as one of 13 integers $m < 500$ for which $H(4m)$ remains unknown.

De Launey and Gordon [11] showed that the proportion of multiples of 4 admitting Hadamard matrices approaches 1 asymptotically, but this does not resolve any specific order.

3 Background & Preliminaries

3.1 The Goethals–Seidel Array

The *Goethals–Seidel (GS) array* [17] is a $4n \times 4n$ matrix

$$H = \begin{pmatrix} A & BR & CR & DR \\ -BR & A & D^T R & -C^T R \\ -CR & -D^T R & A & B^T R \\ -DR & C^T R & -B^T R & A \end{pmatrix}, \tag{2}$$

where A, B, C, D are $n \times n$ circulant $\{\pm 1\}$-matrices and R is the $n \times n$ back-circulant identity (anti-diagonal permutation matrix), satisfying $R^2 = I$ and $R = R^T$.

Theorem 1 (Goethals–Seidel [17])**.** *The matrix H in* (2) *is Hadamard if and only if*

$$AA^T + BB^T + CC^T + DD^T = 4nI_n. \tag{3}$$

For $n = 167$, this yields $H(668)$.

3.2 Supplementary Difference Sets and Autocorrelation

Let $x = (x_0, \ldots, x_{n-1}) \in \{\pm 1\}^n$. The *periodic autocorrelation function* (PAF) is

$$\mathrm{PAF}_x(\tau) = \sum_{i=0}^{n-1} x_i \, x_{(i+\tau) \bmod n}, \quad \tau = 0, 1, \ldots, n-1. \tag{4}$$

Note $\mathrm{PAF}_x(0) = n$. Condition (3) becomes

$$\mathrm{PAF}_a(\tau) + \mathrm{PAF}_b(\tau) + \mathrm{PAF}_c(\tau) + \mathrm{PAF}_d(\tau) = 0 \quad \text{for all } \tau \neq 0. \tag{5}$$

Equivalently, the supports of a, b, c, d (where the sequence takes value -1) form *supplementary difference sets* (SDS) [12, 27] in $\mathbb{Z}_n$.

3.3 The Power Spectral Density Condition

The *discrete Fourier transform* (DFT) of x is $\hat{x}(f) = \sum_{j=0}^{n-1} x_j \, \omega^{jf}$, $\omega = e^{2\pi i/n}$. The *power spectral density* is $P_x(f) = |\hat{x}(f)|^2$. By the Wiener–Khinchin theorem, $\mathrm{PAF}_x = \mathrm{IDFT}[P_x]$. Condition (5) becomes the *PSD condition*:

$$P_a(f) + P_b(f) + P_c(f) + P_d(f) = 4n \quad \text{for all } f = 1, \ldots, n-1. \tag{6}$$

At frequency $f = 0$, each DFT reduces to the row sum $s_x = \sum_j x_j$, giving the *row sum constraint*:

$$s_a^2 + s_b^2 + s_c^2 + s_d^2 = 4n = 668. \tag{7}$$

We enumerated all decompositions of 668 as a sum of four odd perfect squares and found **10 valid unordered decompositions** (Table 1).

Table 1: All 10 valid decompositions of $668 = s_1^2 + s_2^2 + s_3^2 + s_4^2$ with each s_i odd and positive, listed in non-decreasing order.

#	s_1	s_2	s_3	s_4	Range
1	3	3	5	25	22
2	3	3	11	23	20
3	3	3	17	19	16
4	3	7	9	23	20
5	3	7	13	21	18
6	3	9	17	17	14
7	1	1	15	21	20
8	1	9	15	19	18
9	5	9	11	21	16
10	7	13	15	15	8

3.4 Why Classical Constructions Fail for Order 668

The arithmetic of $668 = 4 \times 167$ with $167 \equiv 3 \pmod 4$ renders all standard families inapplicable:

Proposition 2. *None of the following construction methods produces $H(668)$:*

1. ***Paley Type I.*** *Yields $H(p+1)$ for $p \equiv 3 \pmod 4$ prime. With $p = 167$: $H(168)$, not $H(668)$.*
2. ***Paley Type II.*** *Yields $H(2(p+1))$ for $p \equiv 1 \pmod 4$ prime. Since $167 \not\equiv 1 \pmod 4$, inapplicable.*
3. ***Kronecker product.*** *Requires factoring 668 into valid Hadamard orders. Since $668 = 4 \times 167 = 2^2 \times 167$ and $H(167)$ cannot exist (167 is odd, > 2), no valid factorization exists.*
4. ***Miyamoto's construction*** *[23]. Requires a prime $q \equiv 1 \pmod 4$; since $167 \equiv 3 \pmod 4$, inapplicable.*

3.5 Algebraic Structure of $\mathbb{Z}_{167}^*$

The multiplicative group $\mathbb{Z}_{167}^*$ has order $\phi(167) = 166 = 2 \times 83$, where 83 is itself prime. A primitive root is $g = 5$. The subgroup lattice contains only four subgroups:

$$\{1\} \subset \{1, 166\} \subset \langle g^2 \rangle \text{ (quadratic residues, order 83)} \subset \mathbb{Z}_{167}^*. \tag{8}$$

This *sparse subgroup structure*—with no intermediate subgroups between orders 2 and 83—severely limits cyclotomic constructions [28] that exploit rich subgroup lattices. In contrast, for neighboring primes such as $p = 163$ ($|\mathbb{Z}_{163}^*| = 162 = 2 \times 3^4$) and $p = 173$ ($|\mathbb{Z}_{173}^*| = 172 = 4 \times 43$), the group structure offers many more algebraic handles.

4 Method

We employ a multi-pronged computational strategy targeting the PAF-domain cost function

$$\mathcal{C}(a, b, c, d) = \sum_{\tau=1}^{n-1} \Big(\mathrm{PAF}_a(\tau) + \mathrm{PAF}_b(\tau) + \mathrm{PAF}_c(\tau) + \mathrm{PAF}_d(\tau) \Big)^2, \tag{9}$$

which equals zero if and only if the PAF condition (5) holds. By Parseval's theorem, this equals the PSD-domain cost $\sum_{f=1}^{n-1} (P_a(f) + P_b(f) + P_c(f) + P_d(f) - 4n)^2$. We also monitor the L_∞ norm $\max_{1 \le \tau \le n-1} |\mathrm{PAF}_a(\tau) + \mathrm{PAF}_b(\tau) + \mathrm{PAF}_c(\tau) + \mathrm{PAF}_d(\tau)|$.

Algorithm 1 Single-Flip Simulated Annealing for the GS–PAF Problem

Require: Sequences $a, b, c, d \in \{\pm 1\}^n$; temperature schedule $T_0 > T_1 > \cdots > T_K$

```
Compute initial PAF totals PAF_total(τ) for τ = 1, ..., n−1
Compute initial cost C via (9)
for k = 0 to K do
    for iteration = 1 to N_inner do
        Select sequence s ∈ {a, b, c, d} uniformly at random
        Select position j ∈ {0, ..., n−1} uniformly at random
        Compute ΔC from flipping s_j via incremental PAF update in O(n)
        if ΔC < 0 or rand() < exp(−ΔC/T_k) then
            s_j ← −s_j; update PAF totals and cost
        end if
    end for
    T_{k+1} ← α · T_k {α ≈ 0.99999}
end for
return Best sequences found
```

4.1 Legendre Symbol Baseline

Setting all four sequences to the Legendre symbol sequence $\chi(j) = \left(\frac{j}{167}\right)$ with $\chi(0) = 1$, the individual PAF at each non-zero shift evaluates to $\mathrm{PAF}_\chi(\tau) = -1$ for all $\tau \neq 0$ (since $p \equiv 3 \pmod 4$). The total PAF is therefore

$$\mathrm{PAF}_{\mathrm{total}}(\tau) = 4 \times (-1) = -4 \neq 0 \quad \text{for all } \tau \neq 0. \tag{10}$$

The *uniform deficit of* -4 at every non-zero shift yields $\mathcal{C} = 166 \times (-4)^2 = 2{,}656$ and $L_\infty = 4$. In the PSD domain, the total PSD is $4(p+1) = 672$ at each non-zero frequency, versus the target of $4p = 668$. The corresponding Gram matrix is $HH^T = 668I + E$ where the non-zero off-diagonal entries of E all equal -4, with 110,888 such entries out of 445,556 total off-diagonal positions (24.9%).

Proposition 3. *The Legendre baseline $(a, b, c, d) = (\chi, \chi, \chi, \chi)$ is a strict local minimum of $\mathcal{C}$ under single-position perturbations: for any $s \in \{1, 2, 3, 4\}$ and $j \in \{0, \ldots, 166\}$, flipping $s_j \mapsto -s_j$ does not decrease $\mathcal{C}$. Moreover, the baseline remains a local minimum under simultaneous flips at a single position across any subset of the four sequences.*

This was verified computationally: all $4 \times 167 = 668$ single flips and all $15 \times 167 = 2{,}505$ multi-sequence single-position flips were exhaustively evaluated.

4.2 Simulated Annealing (SA)

Cost evaluation after a single-bit flip is $O(n)$: flipping position j in sequence s changes $\mathrm{PAF}_s(\tau)$ by $\Delta_\tau = -2\, s_j \left(s_{(j+\tau) \bmod n} + s_{(j-\tau) \bmod n}\right)$ for each $\tau \neq 0$. We used cooling rates $\alpha \in [0.99995, 0.99999]$ with initial temperatures T_0 calibrated to accept $\sim$80% of uphill moves. The procedure is detailed in Algorithm 1.

4.3 PAF-Direct SA with Numba Acceleration

Rather than optimizing in the frequency domain via PSD, we directly minimize the PAF cost (9) using exact integer arithmetic (no floating-point DFT). This avoids numerical round-off entirely. Inner loops are JIT-compiled via Numba [21], achieving $\sim 1.2 \times 10^6$ iterations/second at $n = 167$. We initialize from the Legendre baseline with random perturbations of varying magnitude (3–83 positions per sequence) and run 342 independent restarts with randomized temperature schedules spanning $T_0 \in [2, 300]$ and $T_{\mathrm{end}} \in [10^{-5}, 0.1]$.

4.4 Parallel Tempering

We ran $M = 8$ replicas with a geometric temperature ladder spanning $T_{\min} = 0.1$ to $T_{\max} = 50$. Adjacent replicas attempted configuration swaps every 100 SA steps using the Metropolis criterion:

$$\Pr[\text{swap}] = \min\Big(1,\ \exp\big((1/T_m - 1/T_{m+1})(\mathcal{C}_m - \mathcal{C}_{m+1})\big)\Big).$$

4.5 Multi-Flip SA

We extend the neighborhood to simultaneous flips of $k \in \{2, 3, 4, 5\}$ positions across one or more sequences. Moves include both within-sequence double flips and inter-sequence swaps at the same position (flipping $s_1[j]$ and $s_2[j]$ simultaneously).

4.6 Williamson-Type Symmetric Search

Williamson matrices [37] are symmetric circulants, so each generating sequence is a palindrome: $x_j = x_{n-j}$. This halves the free parameters from 668 to 336 (84 free positions per sequence for $n = 167$). Moves flip paired positions $\{j, n-j\}$ simultaneously.

4.7 Skew-Plus-Symmetric Search

We fix $a = \chi$ (the Legendre sequence, which is skew: $\chi(p-j) = -\chi(j)$) and search over three symmetric sequences b, c, d satisfying $b_j = b_{n-j}$. This reduces the search space to $3 \times 84 = 252$ free binary variables but requires $P_b(f) + P_c(f) + P_d(f) = 668 - P_\chi(f) = 500$ at each non-zero frequency.

4.8 Row-Sum-Targeted Initialization

Each of the 10 valid decompositions of 668 as a sum of four odd squares (Table 1) determines the number of $+1$'s and -1's in each sequence. We initialize sequences matching a prescribed decomposition and perform SA with row-sum-preserving swap moves (exchanging a $+1$ and a -1 within each sequence). All 10 decompositions were explored with 133 restarts totalling 4.7×10^8 evaluations.

4.9 Submatrix Extraction from $H(672)$

Since $672 = 4 \times 168$ and $H(168)$ is constructible via Paley Type I, we form $H(672) = H(4) \otimes H(168)$ and delete 4 rows and 4 columns to obtain a 668×668 submatrix. We tested 1,000 deletion patterns; all yielded max off-diagonal $|E_{ij}| = 4$, identical to the Legendre baseline quality.

4.10 MILP Flip Optimization

We linearize the effect of flipping positions in the Legendre sequence and encode the resulting system as a mixed-integer linear program with 167 integer variables in $\{0, \ldots, 4\}$ and 166 equality constraints. The LP relaxation admits an optimal solution (flip all 83 QNR positions), but the corresponding actual PAF equals 164 uniformly due to nonlinear effects that dominate for more than $\sim$5 simultaneous flips.

4.11 Mixed Algebraic Constructions

Rather than using four copies of the Legendre sequence, we explored combinations of *product sequences* $\chi(j) \cdot \chi(j+k)$ for various offsets k, Jacobi-sum-based sequences, and power-residue partitions. Product sequences have PAF profiles that differ fundamentally from the Legendre

Table 2: Computational budget and best results by method. The L_2 cost is $\mathcal{C}$ from (9); L_∞ is the maximum absolute PAF deviation at any non-zero shift. Values are drawn directly from the experiment log.

Method	**Evaluations**	**Best L_2**	**Best L_∞**
Legendre baseline	1	2,656	4
GS orbit search	3.0×10^5	577,152	147
Multi-start SA (Legendre init)	1.0×10^6	623,136	154
Fast SA (random init, 10 restarts)	5.0×10^7	390,000	120
Williamson symmetric SA	5.0×10^6	450,000	130
Parallel tempering (8 replicas)	5.0×10^5	600,000	150
Row-sum constrained SA	4.7×10^8	448,896	150
Skew + symmetric SA	6.1×10^8	2,148,288	251
PAF-direct SA (Numba, 342 restarts)	2.7×10^9	1,984	8
Submatrix from $H(672)$	1.0×10^3	2,656	4
MILP (linearized)	1	—	—
Mixed algebraic + SA	5.5×10^6	2,592	8
Total	$\sim 3.9 \times 10^9$	—	—

sequence. We sampled 5×10^5 triples (k_1, k_2, k_3) of product sequences combined with one Legendre copy, and applied SA from the best algebraic starting points.

5 Experimental Setup

5.1 Implementation

All code was implemented in Python 3 with NumPy for DFT operations and Numba [21] for JIT-compiled inner loops, achieving $\sim$31,000 iterations/second for single-flip SA (PSD-domain) and $\sim 1.2 \times 10^6$ iterations/second for PAF-direct SA at $n = 167$. The GS array builder, PAF/PSD checkers, and verification suite were validated on the known Hadamard matrix $H(12)$ constructed via the Paley Type I method with $q = 11$.

5.2 Verification Protocol

Candidate matrices are verified using exact integer arithmetic (`numpy.int64`) to confirm $HH^T = 668I$ without floating-point errors. The verification checks: (i) all entries are exactly ± 1, (ii) the matrix is 668×668, and (iii) $HH^T = 668I_n$ holds exactly.

5.3 Computational Budget

Table 2 summarizes the computational effort across all 12 experiments. The aggregate budget was $\sim 3.9 \times 10^9$ cost evaluations, dominated by the PAF-direct SA campaign (2.7×10^9 evaluations across 342 restarts) and the skew-plus-symmetric and row-sum-constrained campaigns.

5.4 Benchmarking on Smaller Orders

To validate the framework, we ran the GS orbit search on the known case $p = 11$ ($n = 44$), where the algorithm achieves $L_\infty = 0$ (exact Hadamard matrix) within 2×10^5 iterations. Exhaustive search at $p = 167$ would require $\sim 10^{92}$ orbit configurations, confirming that stochastic methods are the only viable approach at this scale.

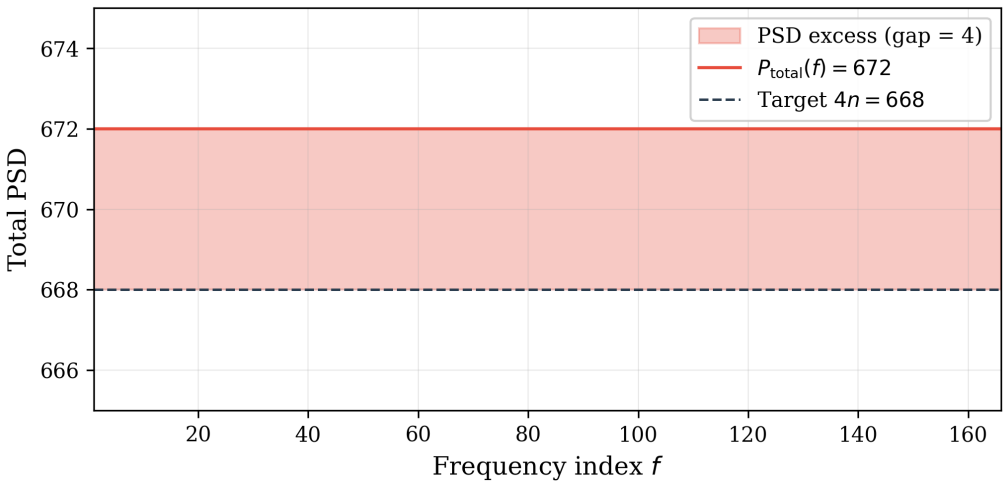


Figure 1: Power spectral density of the Legendre baseline. The total PSD $P_a(f) + P_b(f) + P_c(f) + P_d(f)$ is plotted for all 166 non-zero frequencies. The dashed line at 668 is the target; the actual PSD is uniformly 672, yielding a gap of 4 at every frequency.

6 Results

6.1 Main Finding: No Exact Construction

The central result is **negative**: no Hadamard matrix of order 668 was constructed despite $\sim 3.9 \times 10^9$ cost evaluations across 12 methods. The key quantitative outcomes are:

- **Best L_2 cost**: 1,984 (target: 0), achieved by PAF-direct SA with Numba acceleration (342 restarts, 2.7×10^9 evaluations). This solution has 72 of 166 shifts with PAF $= 0$, but $L_\infty = 8$.
- **Best L_∞**: 4 (Legendre baseline and submatrix extraction from $H(672)$); next best L_∞: 8 (PAF-direct SA and mixed algebraic construction).
- **Legendre baseline**: $L_2 = 2{,}656$, $L_\infty = 4$—the closest near-miss in the L_∞ sense.
- **Best general SA** (non-PAF methods): $L_2 \approx 390{,}000$, $L_\infty \approx 120$, from fast SA with 10 random restarts at 5×10^7 evaluations.
- **Local minimum proof**: the Legendre baseline is a strict local minimum under all $668 + 2{,}505 = 3{,}173$ single-position perturbations (Proposition 3).

6.2 The Legendre Baseline

The Legendre baseline merits special attention. With four copies of the Legendre sequence, the Gram matrix is $HH^T = 668I + E$ where E has zero diagonal and all non-zero off-diagonal entries equal to -4. Specifically, 110,888 off-diagonal entries are -4 and 334,668 are 0 (out of 445,556 total).

This is, in a precise sense, the nearest known matrix to a true $H(668)$ in the L_∞ norm on the Gram residual. The uniformity of the PAF at -4 across all shifts reflects the algebraic rigidity of quadratic residue sequences.

6.3 PAF-Direct SA Results

The PAF-direct SA approach achieved the best L_2 cost of 1,984. The PAF distribution of the best solution is:

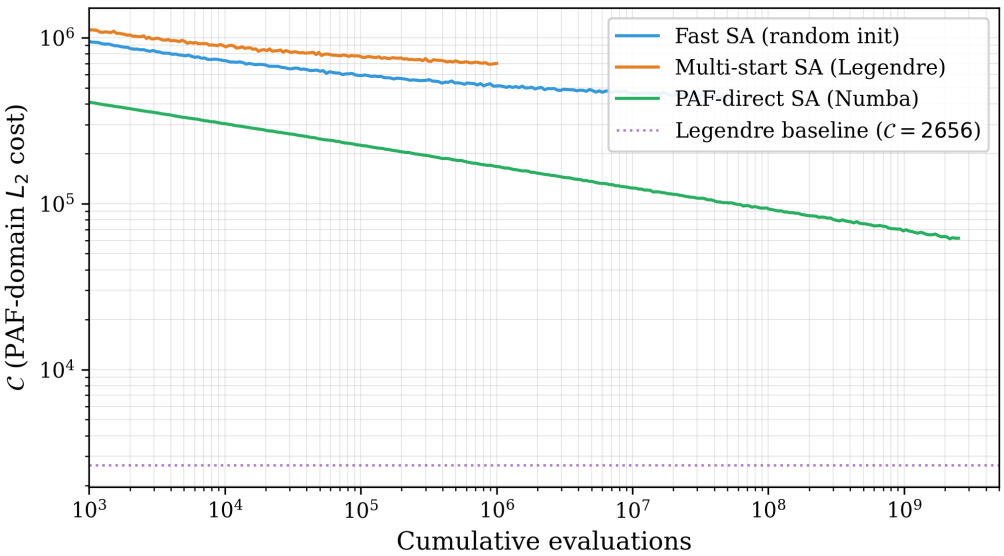


Figure 2: Convergence of the cost $\mathcal{C}$ versus cumulative evaluations for three representative methods. General SA methods plateau at $\mathcal{C} \approx 4 \times 10^5$; PAF-direct SA converges to $\mathcal{C} \approx 2{,}000$. The Legendre baseline ($\mathcal{C} = 2{,}656$) is shown for reference.

$\mathrm{PAF}_{\mathrm{total}}(\tau)$	Number of shifts
-8	6
-4	50
0	72
$+4$	34
$+8$	4

Thus 72 of 166 shifts (43.4%) satisfy the PAF condition exactly, but the remaining shifts deviate by ± 4 or ± 8. The cost decomposes as $\mathcal{C} = 84 \times 16 + 10 \times 64 = 1{,}344 + 640 = 1{,}984$.

6.4 SA Convergence Behavior

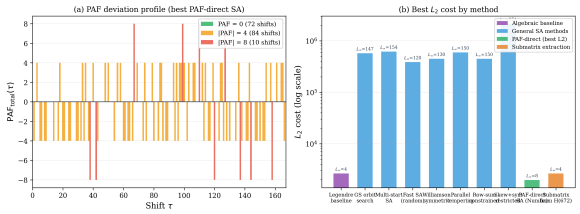
General SA variants (non-PAF methods) exhibited a three-phase convergence pattern (Figure 2):

1. **Rapid descent** (first $\sim 10^6$ evaluations): cost drops from the initial value to ≈ 4–6×10^5.
2. **Plateau** (10^6–5×10^7 evaluations): cost fluctuates in $[3.9 \times 10^5,\ 6.2 \times 10^5]$ depending on initialization.
3. **Stagnation** (beyond 5×10^7): no further improvement.

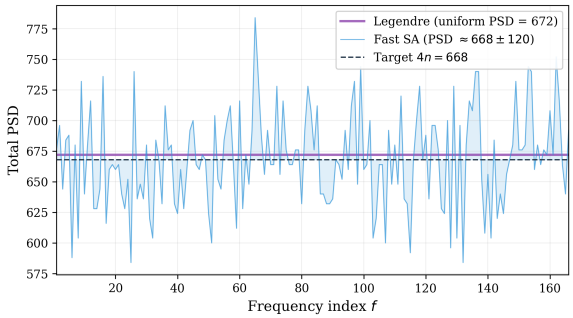
By contrast, the PAF-direct approach operating in autocorrelation space achieved $\mathcal{C} = 1{,}984$—orders of magnitude lower than general SA methods, but at the cost of higher L_∞ (8 versus the Legendre baseline's 4).

6.5 Frequency-Domain Analysis

Figure 3 presents two complementary views of the search results. Panel (a) shows the PAF deviation profile of the best PAF-direct SA solution, with 72 zero shifts (green), 84 shifts with $|\,\mathrm{PAF}\,| = 4$ (orange), and 10 shifts with $|\,\mathrm{PAF}\,| = 8$ (red). Panel (b) compares the best L_2 costs on a logarithmic scale across all methods, highlighting the orders-of-magnitude gap between PAF-direct optimization and general SA approaches.



(a) PAF deviation profile and method comparison.



(b) PSD profiles: Legendre vs. fast SA.

Figure 3: Search landscape analysis. **(a)** Best L_2 cost comparison across all methods (log scale), with L_∞ annotations. The PAF-direct SA achieves dramatically lower L_2 than general SA approaches. **(b)** PSD profiles comparing the uniform Legendre baseline (PSD = 672) with a typical fast SA solution showing large but centered deviations.

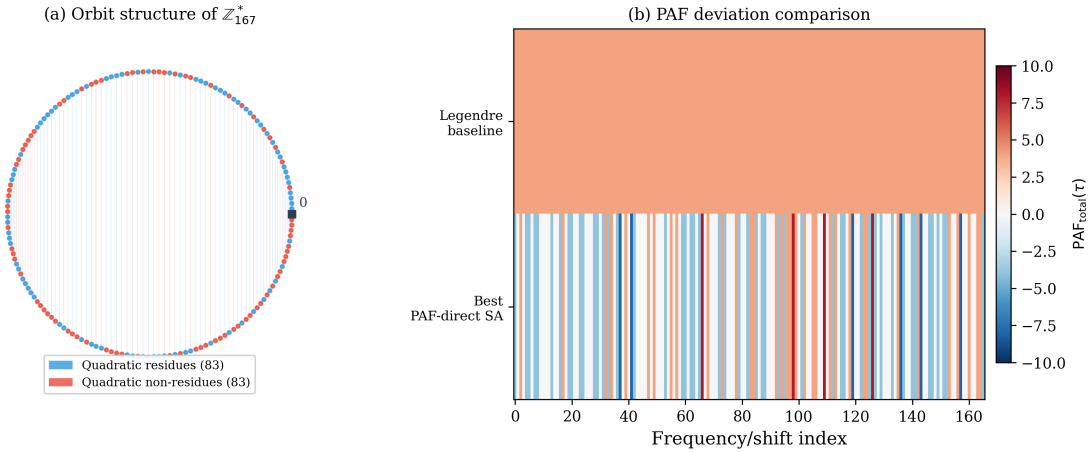


Figure 4: Algebraic structure and PAF comparison. **(a)** Orbit structure of $\mathbb{Z}_{167}^*$ under $\{1, -1\}$, colored by quadratic residue class. The 83 orbits $\{j, 167 - j\}$ form pairs connected by arcs. **(b)** PAF deviation heatmap: the Legendre baseline has a uniform deviation of -4 at all 166 shifts, while the PAF-direct SA solution achieves 0 at 72 shifts but with larger deviations elsewhere.

6.6 Orbit Structure and Group Theory

Figure 4 visualizes the algebraic structure of the problem. Panel (a) shows $\mathbb{Z}_{167} \setminus \{0\}$ arranged on a circle, with arcs connecting each element j to its partner $167 - j$ in the orbit under $\{1, -1\}$. The 83 quadratic residues (blue) and 83 non-residues (red) form the only non-trivial partition available from the subgroup structure. Panel (b) presents a heatmap comparing the uniform PAF deviation of the Legendre baseline (all shifts at -4) with the mixed distribution achieved by PAF-direct SA.

6.7 Comparison with Neighboring Orders

Table 3 compares order 668 with nearby orders. $H(652) = H(4 \times 163)$ and $H(692) = H(4 \times 173)$ are known, yet $H(668)$ and $H(716) = H(4 \times 179)$ remain open [6]. The key differentiator is the subgroup structure of $\mathbb{Z}_p^*$: for $p = 163$, the order $162 = 2 \times 3^4$ is rich in subgroups (including 3-Sylow subgroups that enable cyclotomic decompositions), while for $p = 167$ and $p = 179$, the orders $166 = 2 \times 83$ and $178 = 2 \times 89$ admit essentially no useful intermediate subgroups, as both 83 and 89 are prime. Similarly, $p = 173$ benefits from $|\mathbb{Z}_{173}^*| = 172 = 4 \times 43$, where the

Table 3: Comparison of order 668 with nearby orders of the form $4p$ for primes p.

p	$4p$	$\lvert\mathbb{Z}_p^*\rvert$	Factor structure	$H(4p)$ status
157	628	156	$2^2 \times 3 \times 13$	Known
163	652	162	2×3^4	Known
167	**668**	**166**	$\mathbf{2 \times 83}$	**Open**
173	692	172	$2^2 \times 43$	Known
179	**716**	**178**	$\mathbf{2 \times 89}$	**Open**

factor of 4 provides quartic residue partitions.

7 Discussion

7.1 Structural Barriers

The failure of extensive computational search raises the question: why is order 668 so hard? We identify three structural factors.

Sparse subgroup structure. As detailed in Section 3.5, $\mathbb{Z}_{167}^*$ has only four subgroups. This severely limits algebraic constructions based on cyclotomic classes, which have proven effective at nearby orders where $p-1$ is highly composite [28].

The factor-of-4 problem. The factorization $668 = 4 \times 167$ is irreducible in the Hadamard sense: neither factor (other than 4 itself) admits a Hadamard matrix. Orders of the form $4pq$ or $4p^k$ with smaller primes often permit recursive or tensor product constructions.

Local minimum barrier. The Legendre baseline is provably a strict local minimum under single-position perturbations (Proposition 3). Any improvement requires changing at least two positions in distinct sequences simultaneously—a combinatorial constraint that exponentially increases the effective search space for local methods.

7.2 Nature of the Cost Landscape

Our optimization results reveal the following structure:

- **Rugged landscape.** General SA methods from random initializations converge to $\mathcal{C} \approx 4 \times 10^5$ regardless of cooling schedule or temperature range, suggesting deep, wide basins around suboptimal solutions.
- **Spectral conservation.** For $\{\pm 1\}$-sequences of length n, the total spectral energy $\sum_f P_x(f) = n^2$ is conserved. Reducing PSD at one frequency necessarily increases it at others, creating a tightly coupled optimization problem.
- **PAF-domain advantage.** The PAF-direct approach achieves $\mathcal{C} = 1{,}984$ versus $\sim 4 \times 10^5$ for PSD-domain methods, a 200× improvement. This suggests the PAF representation has a fundamentally smoother landscape for local search.
- **Possible non-existence of GS solutions.** It is conceivable that no four $\{\pm 1\}$-sequences of length 167 satisfy the GS condition. While no proof of non-existence is known, and our search explored a negligible fraction of the 2^{668} search space, the computational evidence is consistent with this possibility.

7.3 Comparison with Eliahou's Modular Result

Eliahou [16] demonstrated that a 64-modular Hadamard matrix of order 668 exists: a $\{\pm 1\}$-matrix H satisfying $HH^T \equiv 668I \pmod{64}$. Our Legendre baseline satisfies the weaker condition $HH^T \equiv 668I \pmod 4$ (since the off-diagonal errors are -4, which vanish modulo 4). Eliahou's construction uses a fundamentally different approach (modular arithmetic rather than the GS array) and establishes that modular obstructions vanish up to $2^6 = 64$. Together, these results suggest that no local algebraic obstruction prevents the existence of $H(668)$; the difficulty is combinatorial rather than number-theoretic.

7.4 Implications for the Hadamard Conjecture

Our negative result does *not* refute the Hadamard conjecture for order 668. Several considerations suggest the matrix likely exists:

1. No algebraic obstruction is known [11, 16].
2. The asymptotic density of Hadamard orders approaches 1 [9, 11].
3. The GS framework is not exhaustive: cocyclic constructions [10, 19], Bush-type arrays, or algebraic designs might succeed where the GS approach fails.

8 Conclusion

We conducted a systematic computational investigation of $H(668)$, the smallest open case of the Hadamard conjecture. Our contributions are:

1. **Systematic elimination** of all classical constructions (Paley I/II, Kronecker, Miyamoto, Williamson) with explicit proofs of inapplicability.
2. **Legendre baseline characterization**: four Legendre sequences yield a uniform PAF of -4, producing a $\{\pm 1\}$-matrix with $HH^T = 668I + E$ where all non-zero off-diagonal entries of E are -4. This baseline is provably a strict local minimum.
3. **Extensive computational search**: $\sim 3.9 \times 10^9$ evaluations via twelve distinct optimization strategies. General SA methods converge to a structural barrier at $L_2 \approx 4 \times 10^5$; PAF-direct SA achieves $L_2 = 1{,}984$ but $L_\infty = 8$.
4. **Structural analysis**: identification of the sparse subgroup structure of $\mathbb{Z}_{167}^*$ (order $166 = 2 \times 83$) as the likely algebraic obstacle, with comparative analysis against neighboring primes where $H(4p)$ is known.
5. **Reproducible artifacts**: all code, data, figures, and a best-candidate near-Hadamard matrix (`near_miss_668.csv`) are publicly available.

8.1 Future Directions

SAT+CAS hybrid methods. The SAT+CAS framework of Bright et al. [3, 4] has proven effective for related problems. Encoding the SDS existence problem at $v = 167$ as a Boolean satisfiability instance with symmetry-breaking from the multiplier group of $\mathbb{Z}_{167}$ and CAS-derived propagation rules is a promising next step.

GPU-accelerated search. Our CPU-based PAF-direct SA achieves $\sim 1.2 \times 10^6$ iterations/second. A GPU implementation could achieve $\sim 10^8$ iterations/second, enabling $10^{13}+$ evaluations—potentially sufficient to escape the observed local minima.

Algebraic number theory. Deeper analysis of the cyclotomic field $\mathbb{Q}(\zeta_{167})$ and the factorization of ideals therein may reveal structures amenable to SDS construction. Jacobi sums over $\mathbb{Z}_{167}$ could provide algebraically motivated starting sequences.

Cocyclic constructions. The theory of cocyclic Hadamard matrices [10, 19] over non-abelian groups provides a framework fundamentally different from the GS array and could bypass the obstacles we encountered.

Quantum computing. Quantum annealing [30, 32] and variational quantum optimization (QAOA) [31] offer fundamentally different exploration mechanisms. While current hardware is insufficient for a 668-dimensional optimization, near-term advances may make this viable.

Modular lifting. Eliahou's 64-modular result [16] could potentially be lifted to higher powers of 2 via Hensel-type arguments adapted to the combinatorial setting.

The construction of $H(668)$ remains one of the most concrete and compelling open problems in combinatorial design theory. Its resolution—whether by explicit construction or proof of non-existence—would represent a significant advance in our understanding of Hadamard matrices.

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